

WEICHANG, China

WMO: 543110

Lat: 41.9605N

Lon: 117.7661E

Elev: 894

StdP: 91.04

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-20.2	-31.4	0.2	-17.6	-29.2	0.3	-16.3	8.2	-12.5	7.4	-12.6	2.3	0	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.6	18.5	29.0	18.4	27.7	17.8	22.2	26.6	21.2	25.7	20.2	24.7	3.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	17.3	24.8	19.8	16.2	23.8	18.7	15.1	22.9	70.1	26.6	66.3	25.6	62.6	24.8	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.2	5.3	DB	-24.9	33.7	2.7	1.8	-26.8	35.0	-28.4	36.1	-29.9	37.1	-31.9	38.5	(4)
(5)			WB	-25.3	23.5	2.6	1.3	-27.1	24.4	-28.6	25.2	-30.1	25.9	-32.0	26.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-11.8	-8.0	-0.7	7.8	14.2	18.8	21.8	20.1	14.6	6.8	-2.6	-10.2	(6)
(7)		DBStd	12.39	3.90	4.63	4.95	4.53	3.66	2.93	2.31	2.53	3.47	4.05	4.92	3.91	(7)
(8)		HDD10.0	2744	677	505	333	95	8	0	0	0	6	114	378	627	(8)
(9)		HDD18.3	4752	936	738	591	318	136	28	2	12	119	358	628	886	(9)
(10)		CDD10.0	1268	0	0	1	28	137	266	366	314	142	14	0	0	(10)
(11)		CDD18.3	234	0	0	0	1	7	43	110	67	6	0	0	0	(11)
(12)		CDH23.3	2110	0	0	0	12	173	482	848	509	84	2	0	0	(12)
(13)		CDH26.7	528	0	0	0	1	36	131	248	104	9	0	0	0	(13)
(14)	Wind	WSAvg	2.1	2.3	2.2	2.5	2.6	2.5	2.0	1.8	1.7	1.9	2.0	2.0	2.2	(14)
(15)	Precipitation	PrecAvg	422	2	4	7	15	35	67	129	96	46	17	4	1	(15)
(16)		PrecMax	648	11	11	31	50	81	111	331	223	88	61	14	9	(16)
(17)		PrecMin	249	0	0	0	0	7	17	19	27	4	0	0	0	(17)
(18)		PrecStd	94	3	3	7	11	21	24	68	54	22	13	3	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.3	7.7	17.9	25.1	30.4	32.3	33.6	31.3	28.0	21.8	12.5	3.0	(19)
MCWB			-3.3	1.2	6.3	11.6	15.1	16.4	20.8	19.6	15.5	11.8	4.7	-1.2	(20)	
2%		DB	-1.3	4.9	13.8	21.9	27.3	29.8	30.8	29.2	25.7	18.7	9.6	0.2	(21)	
		MCWB	-5.2	-1.0	4.6	9.7	13.6	16.9	19.5	18.8	15.6	10.1	3.6	-3.8	(22)	
5%		DB	-3.1	2.8	10.8	19.6	25.2	28.1	29.2	27.7	23.8	16.5	7.4	-1.8	(23)	
		MCWB	-6.5	-2.2	2.8	8.9	12.7	16.4	19.8	18.7	14.6	9.0	2.0	-5.3	(24)	
10%		DB	-4.9	0.4	8.1	17.1	23.0	26.3	27.8	26.2	21.9	14.4	4.9	-3.5	(25)	
		MCWB	-7.8	-3.8	1.1	7.6	11.4	15.9	19.3	18.3	14.0	7.8	0.4	-6.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.9	1.7	7.6	14.0	17.3	20.5	23.7	23.1	19.4	13.7	5.9	-1.3	(27)
MCDWB			0.8	5.7	16.5	21.4	25.9	26.8	28.9	27.4	24.7	18.8	10.4	3.1	(28)	
2%		WB	-5.0	-0.8	5.1	11.5	15.4	19.2	22.6	21.9	17.5	11.8	4.0	-3.6	(29)	
		MCDWB	-1.6	4.5	12.8	19.9	23.4	25.1	27.5	25.8	22.5	16.6	8.8	-0.3	(30)	
5%		WB	-6.4	-2.2	3.2	9.7	14.2	18.4	21.7	20.9	16.3	10.3	2.4	-5.0	(31)	
		MCDWB	-3.3	2.5	10.2	17.4	21.6	24.2	26.3	24.9	21.2	15.0	6.9	-1.8	(32)	
10%		WB	-7.8	-3.7	1.5	8.2	13.0	17.5	20.9	19.8	15.1	8.7	0.7	-6.4	(33)	
		MCDWB	-5.0	0.4	7.7	16.1	20.5	23.1	25.4	23.9	19.9	13.1	4.7	-3.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.1	12.5	12.7	13.4	14.0	12.3	10.7	11.3	12.6	12.3	11.2	10.6	(35)
MCDBR			13.5	15.3	16.7	17.8	18.7	16.1	13.7	13.8	15.6	15.5	13.7	12.7	(36)	
MCWBR			10.2	10.5	9.7	8.8	8.0	6.0	5.0	5.5	6.8	8.2	8.5	9.5	(37)	
5% WB		MCDWB	13.3	14.7	15.6	15.0	14.6	11.5	10.0	10.1	11.9	13.1	13.1	12.6	(38)	
		MCWBR	10.3	10.2	9.3	7.9	6.8	5.1	4.6	4.9	6.2	7.8	8.4	9.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.294	0.361	0.407	0.477	0.538	0.610	0.573	0.530	0.481	0.425	0.331	0.302	(40)	
taud		2.291	2.086	1.999	1.877	1.775	1.692	1.824	1.903	1.948	2.013	2.226	2.262	(41)		
Ebn at Noon		845	819	825	793	754	701	723	740	743	734	782	794	(42)		
Edh at Noon		87	127	157	191	217	237	206	185	164	136	93	83	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.54	3.51	4.73	5.53	6.14	5.57	5.39	5.26	4.57	3.53	2.56	2.18	(44)	
(45)		RadStd	0.14	0.24	0.30	0.38	0.41	0.49	0.34	0.31	0.27	0.24	0.20	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	-0.60	-1.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (6 neighbors)	N/A	N/A	-1.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page